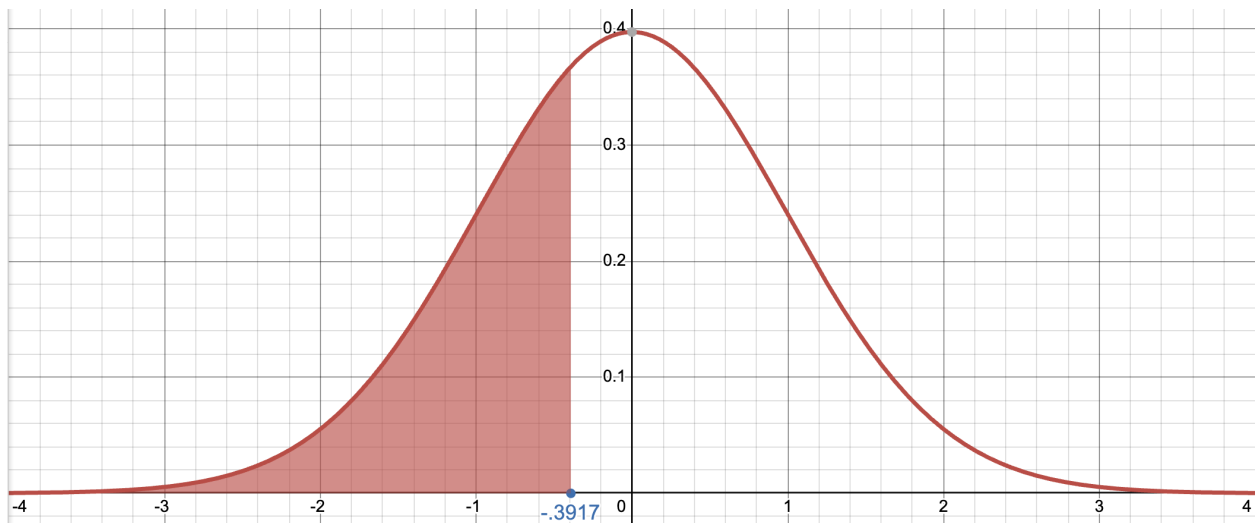


Analysis

Now, we'll perform the two-sample t-test for $\mu_p - \mu_{np}$.

$$\begin{aligned} t &= \frac{\bar{x}_p - \bar{x}_{np}}{\sqrt{\frac{s_p^2}{n_p} + \frac{s_{np}^2}{n_{np}}}} \\ &= \frac{0.381 - 0.398}{\sqrt{\frac{(0.335)^2}{68} + \frac{(0.290)^2}{361}}} \\ &= \frac{-0.017}{0.043397} \\ &= -0.3917 \end{aligned}$$

$$\begin{aligned} p &= 2 \cdot \text{tcdf}\left(\overset{\text{lower}}{-1\text{E}99}, \overset{\text{upper}}{-0.3917}, \overset{\text{df}}{67}\right) \\ &= 2 \cdot 0.3483 \\ &= \boxed{0.6966} \end{aligned}$$



Assuming H_0 is true (Shohei's true mean wOBA is the same when pitching vs not pitching), there is a 0.6966 probability of getting a sample with a mean difference as large or larger than the one in this sample (.017).

Since $p = 0.6966 > \alpha = 0.05$, we fail to reject H_0 . There is not convincing evidence that Shohei Ohtani's true mean wOBA is different when pitching vs not pitching. This conclusion would hold for any common value of α (0.01, 0.05, 0.10). This supports my prediction that Shohei would bat similarly regardless of position.

There is a possible type II error, because we failed to reject H_0 . This would mean that Shohei actually does bat differently when pitching compared to not pitching. Although a type I error is not possible, as we did not reject H_0 , it would mean that Shohei does not bat differently, but we said that he did.

Next, we'll perform a two-sample t-interval for $\mu_p - \mu_{np}$, at confidence level 95%.

$$\begin{aligned}
 \text{CI} &= (\bar{x}_p - \bar{x}_{np}) \pm t^* \sqrt{\frac{s_p^2}{n_p} + \frac{s_{np}^2}{n_{np}}} \\
 &= (\bar{x}_p - \bar{x}_{np}) \pm \text{invT}\left(1 - \alpha/2, \text{df}\right) \sqrt{\frac{s_p^2}{n_p} + \frac{s_{np}^2}{n_{np}}} \\
 &= (0.381 - 0.398) \pm \text{invT}\left(0.975, 67\right) \sqrt{\frac{(0.335)^2}{68} + \frac{(0.290)^2}{361}} \\
 &= -0.017 \pm 1.996 \cdot 0.043397 \\
 &= -0.017 \pm 0.08662 \\
 &= \boxed{(-0.1036, 0.0696)}
 \end{aligned}$$

We are 95% confident that $(-0.1036, 0.0696)$ contains the true mean difference in wOBA when Shohei is pitching vs not pitching.

Since 0 is on the interval, $H_0 : \mu_p = \mu_{np}$ is plausible. This supports the t-test, in which we failed to reject H_0 . The interval gives us a range of plausible values for $\mu_p - \mu_{np}$, showing there is likely a small to zero difference when pitching vs not pitching.